



Multi-scale modeling of solidification and microstructure development in laser keyhole welding process for austenitic stainless steel



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ABSTRACT

A multi-scale model is developed in this paper to investigate the growth of grains and sub-grain dendrites during the molten pool solidification of austenitic stainless steel AISI304 in laser keyhole welding processes. A macro-scale dynamic model is used to predict the fluid flow and heat transfer in the keyhole as well as the molten pool. The generated thermal history is fed to a three-dimensional Cellular Automata model to predict the meso-scale grain growth and to a two-dimensional Cellular Automata–Phase Field model to predict the micro-scale dendrite morphology. The multi-scale model is applied to simulate multiple cases of different welding parameters. The model predictions are validated against experiments and studied further for detailed information. The effects of the welding parameters are discussed based on the numerical and experimental results.

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1. Introduction

Laser keyhole welding is widely utilized in industrial manufacturing owing to its ability to produce deep but narrow weldments with a high efficiency. The final microstructure in the re-solidified region, i.e. the weld zone, is a quality issue of great importance, and its evolution is highly dependent on the transport phenomena and phase change during the welding process [1,2]. A comprehensive understanding of the solidification process is essential for the prediction and control of final microstructure and properties of weldments. While experimentation is usually used as the major approach to investigate this issue, numerical modeling has been increasingly gaining attention.

Extensive efforts have been spent to model the dynamics of keyhole and its surrounding molten pool. During computational modeling of heat transfer and fluid mechanics, special attention has to be paid to the calculation of keyhole shape. Diverse attempts have been made to determine the keyhole shape [3–7], and these fundamental studies have led to the more advanced models that enable the dynamic determination of the keyhole shape to be fully coupled with the computational heat transfer and fluid flow. With the utilization of an extra governing equation, either the Level-set (LS) model [8–11] or Volume-of-Fluid (VOF) model [12–15], these numerical models can provide detailed information regarding the keyhole growth, and the distribution of temperature and flow

velocity in the molten pool can also be acquired. While these models do not account for the microstructure development in the mushy region, they can provide reasonable prediction of the thermal history in the mushy region, which is proved to profoundly affect the resultant microstructure.

The microstructure development in fusion welding and deposition processes has also been studied by various numerical models. The meso-scale Cellular Automata (CA) model [16,17] and the Monte Carlo method [18,19] have been used to predict the grain growth in the weld zone. On the other hand, the sub-grain level dendrite growth has been investigated by the micro-scale CA model and the Phase Field (PF) models. The CA models are efficient tools to predict the dendrite growth in a large area of molten pool, and have successfully provided good predictions of microstructure evolution of binary alloys in welding and deposition processes [20–22]. The PF models are also widely applied to the molten pool solidification of binary alloys [23–26], and a successful application to multi-component alloys in welding processes can also be found in [27]. While the PF model is proved to be more thermodynamically sound, it usually requires much higher computational expense, which limits the simulation areas to be relatively small. Based on the development of the CA and PF models, a novel scheme has been proposed in [28] to integrate the meso-scale CA model with the PF model for the prediction of grain growth of multi-component alloys in casting processes. This scheme was extended in [29] to integrate the micro-scale CA model with the PF model for the prediction of dendrite growth for multi-component alloys. The CA–PF model is capable of producing quantitative prediction

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of dendrite morphology with an affordable computational cost, and has been applied to predict the microstructure of multi-component alloys in laser direct deposition and laser conduction welding processes [30,31].

The microstructure development during the molten pool solidification in a laser keyhole welding process is generally more complicated than those in other fusion welding or deposition processes. First, the formation of the keyhole generates a molten pool of unique shape. The aspect ratio is usually high for the molten pool, and the pool size changes non-linearly along all the three dimensions. The complex shape of the molten pool offers preference to different crystallographic orientations at different locations, and hence the overall grain distribution in the final weld can be rather complex. Second, the laser traveling speed is fast in keyhole welding processes, leading to a high cooling rate in the mushy region. Accordingly, the solidification can be rapid and far from equilibrium, producing fairly fine dendrites on the sub-grain level. Probably due to the physical complexity, little work can be found on the modeling of microstructure development during laser keyhole welding processes, and the goal of this paper is to present a comprehensive prediction of this matter. A multi-scale numerical model has been developed to this end, and the model predictions are validated against experimental results. The effects of the laser welding parameters are discussed based on both the simulation and experiment results. The present work is geared towards austenitic stainless steel AISI304, but the approach can be readily extended to other materials.

2. Model description

The multi-scale model is composed of three sub-models on different scales. On the macro-scale, a multi-phase and multi-physics model is used to simulate the three-dimensional (3D) dynamic evolution of keyhole and molten pool. The model provides information about the molten pool shape as well as the solidification conditions in the mushy region. Such information is first applied to a meso-scale 3D-CA model to simulate the competitive grain growth in the molten pool and predict the grain distribution in the entire final welds. The information is also applied to a micro-scale CA-PF model to predict the sub-grain dendrite growth and final morphology on selected planar sections in the weld zone.

2.1. Macro-scale dynamic model for keyhole and molten pool

A 3D numerical model has been developed to simulate process dynamics in the laser keyhole welding processes with a comprehensive consideration of laser-matter interaction, free surface motion and multi-phase heat transfer and fluid flow. While the detailed description of the model is given in [10,32], the governing equations and boundary conditions are briefly introduced here for the sake of completeness.

The free interface between the condensed region (including liquid and solid phases) and the non-condensed region (including metallic vapor and ambient gases), a part of which being the keyhole wall, is tracked by the solution of the Level-set (LS) equation (Eq. (1)).

$$\frac{\partial \varphi}{\partial t} = - \left(\vec{V} + \frac{M_{evap}}{\rho_v} \vec{N} \right) \cdot \nabla \varphi \quad (1)$$

Here φ is the LS value, which is defined to be zero on the interface of interest. Any point off the interface will have its LS value being the signed distance from this point to the interface, with a positive sign in the non-condensed region and a negative sign in the condensed region. The source term on the right hand side (RHS) of the equation calculates the normal velocity of the interface, which

includes two components. The first one is due to the local liquid convection $\vec{V}$, and the second one is due to the surface recession induced by evaporation mass loss. M_{evap} is the mass evaporation rate, which is calculated by the evaporation kinetics proposed by Knight [33], and ρ_v is the vapor density.

The transport phenomena in both the condensed and non-condensed regions are calculated based on the conservation equations of mass, chemical species, momentum and energy (Eqs. (2)–(5)).

$$\frac{\partial}{\partial t}(\rho) + \nabla \cdot (\rho \vec{V}) = 0 \quad (2)$$

$$\frac{\partial}{\partial t}(\rho Y) + \nabla \cdot (\rho \vec{V} Y) = \nabla \cdot (D \rho \nabla Y) \quad (3)$$

$$\begin{aligned} \frac{\partial}{\partial t}(\rho \vec{V}) + \nabla \cdot (\rho \vec{V} \vec{V}) = & + \nabla \cdot (\mu \nabla \vec{V}) - \nabla p - \frac{\mu}{K} \vec{V} + \rho_r \vec{g} \beta_T (T - T_r) \\ & + \nabla_s T \frac{\partial \gamma}{\partial T} \delta_m(\varphi) \end{aligned} \quad (4)$$

$$\frac{\partial}{\partial t}(\rho h) + \nabla \cdot (\rho \vec{V} h) = \nabla \cdot (k \nabla T) \quad (5)$$

In the above equations, ρ is the mass density, $\vec{V}$ is the velocity, Y is the mass fraction of the metal vapor in the Argon environment, and D is the mass diffusion coefficient of the iron vapor, μ is the viscosity, K is the isotropic permeability expressed by the Kozeny–Carman equation, ρ_r is the reference density at the reference temperature T_r , $\vec{g}$ is the gravitational force vector, β_T is the thermal expansion coefficient, $\nabla_s T \partial \gamma / \partial T$ is the thermo-capillary force, $\delta_m(\varphi)$ is the modified delta function, h is the material enthalpy, and k is the thermal conductivity.

The material enthalpy h is calculated as a function of temperature according to Eq. (6).

$$h_s(T) = C_s T \quad (\text{for solid metal}) \quad (6a)$$

$$h_{mushy}(T) = h_s(T_{solidus}) + L_{melt} f_s \quad (\text{for mushy region}) \quad (6b)$$

$$h_l(T) = h_s(T_{solidus}) + L_{melt} + C_l(T - T_{liquidus}) \quad (\text{for liquid metal}) \quad (6c)$$

$$h_v(T) = h_l(T_{boiling}) + L_{evap} + C_v(T - T_{boiling}) \quad (\text{for vapor metal}) \quad (6d)$$

$$h_{gas}(T) = C_{gas} T \quad (\text{for assisting gas}) \quad (6e)$$

Here C_x is the specific heat for metal solid, liquid, vapor or assisting gas according to the subscript, T_x is the solidus, liquidus or boiling temperature of the metal according to the subscript, and f_s is the volume fraction of the solid metal in the mushy region.

In the vapor region, the material is assumed to be the mixture of the Iron vapor (evaporated from the substrate of stainless steel) and the shielding gas (Argon). The viscosity and thermal conductivity of the mixture are calculated based on the formulation proposed by Wilke [34] and Mathur [35]. In the liquid region, several extra forces (including the dragging force in the mushy region, the thermo-gravitational force in the entire molten pool, and the thermo-capillary force applied on the molten pool surface along the tangent direction) are present. These forces are formulated as extra source terms in Eq. (4) (i.e. the third to the fifth terms on the RHS) and they are active only for the liquid phase.

Note that no source term in the governing equations are related to the evaporation process. This phenomenon is addressed by the boundary conditions on the free interface (the zero-levelset). According to the kinetic theory in [33], there exist sharp jumps of temperature, heat flux, fluid velocity and pressure on the

keyhole wall. Using the notation $[X]$ to denote the jump condition of a physical quantity, X , across the interface, the temperature jump across the interface can be expressed as

$$[T] = T_l - T_v \quad (7)$$

Here T_l and T_v are the temperatures on the liquid side and vapor side of the keyhole wall, which can be calculated according to [33]. The heat flux jump is due to the laser absorption, surface radiation and evaporation, and can be formulated as in Eq. (8).

$$[f] = [k\nabla T + \rho\vec{V}h] = Q_{laser} - \sigma\epsilon(T^4 - T_\infty^4) - m_{evap}L_{evap} \quad (8)$$

On the RHS of this equation, the first term, Q_{laser} , is the heat influx due to the laser absorption on the keyhole wall, which can be evaluated based on the ray-tracing model [10]. The second term is the heat efflux due to the surface radiation, where σ is the Stefan–Boltzmann constant and ϵ is the material emissivity. The third term is the latent heat required for local evaporation, where m_{evap} is the evaporated mass rate and L_{evap} is the latent heat of evaporation. A certain amount of liquid phase will be vaporized, and this produces a sharp increase of V_n , the velocity component along the interface normal, which can be formulated as in Eq. (9) [36].

$$[V_n] = m_{evap} \left[\frac{1}{\rho} \right] \quad (9)$$

A number of forces are acting on the interface to produce a pressure jump as shown in Eq. (10).

$$[P] = \gamma\kappa + P_{plume} \quad (10)$$

Here the term $\gamma\kappa$ is the capillary force on the liquid surface, in which γ and κ are the surface tension coefficient and surface curvature, respectively. The term P_{plume} is the force generated by the vapor plume on the liquid surface and two major mechanisms contribute to this force. First, the evaporation generates a recoil pressure on the local liquid surface, as can be calculated by Eq. (11) [37],

$$P_{recoil} = 0.55P_0 \exp\left(\frac{L_{evap}}{R_v T_{boiling}} \left(1 - \frac{T_{boiling}}{T}\right)\right) \quad (11)$$

in which P_0 is the atmospheric pressure and R_v is the gas constant. Second, the vapor flow may impact the liquid surface, and such an effect is measured by the stagnation pressure of the vapor phase, P_{stag} , which can be calculated by Eq. (12).

$$P_{stag} = P_{static} + \frac{\rho V_n^2}{2} \quad (12)$$

Here P_{static} is the static pressure component of the vapor phase, and $\rho V_n^2/2$ is the dynamic pressure component, with V_n being the normal component of the vapor velocity with respect to the keyhole wall. The plume effect is then evaluated by Eq. (13).

$$P_{plume} = \max(P_{recoil}, P_{stag}) \quad (13)$$

The above equations determine the state of the vapor flow on the keyhole wall, which can be used to calculate the influx of mass, chemical species, momentum and energy of the vapor flow into the non-condensed region. All these flux terms serve as the boundary conditions for the conservation equations (Eqs. (7)–(13)) in the non-condensed region to quantitatively account for the evaporation phenomenon.

With the above formulation, the model is able to simulate the dynamic evolution of keyhole and molten pool, and the predicted thermal history will be used as an input for the predictions of microstructure development. The macro-scale thermal history determines the solidification conditions in the mushy region and hence the grain and dendrite growth on the meso and micro-scale. In the meanwhile, the microstructure development also reversely affects the macro-scale phenomena. First, a certain amount of

solidification latent heat is released in the mushy region to modify the macro-scale temperature distribution, and this phenomenon is implicitly considered by incorporating the latent heat in the material enthalpy (Eq. (6)) when calculating the macro-scale energy conservation equation (Eq. (5)). Second, the dendrites in the mushy region generate a dragging force for the molten material flow, and this effect is accounted for with an additional source term in the macro-scale momentum conservation equation (3rd term on RHS of Eq. (4)). The released latent heat and the dragging force in the mushy region are both dependent on the microstructure development, but such dependences are not considered in the present work.

2.2. Three-dimensional meso-scale Cellular Automata model for grain growth

The grain structure is highly dependent on the shape of the molten pool [38,39]. Since the molten pool shape is rather complex during laser keyhole welding processes, the grain distribution in the weld zone should be strongly 3-dimensional [38]. A numerical model has been developed to simulate this process, and the model has to account for the 3D competitive growth among different grains in order to give reasonable prediction.

The model is applied in a 3D computational domain, which is designed to cover a certain volume of the substrate. The domain is meshed into a number of cubic cells with uniform size, and each cell carries the variables of cell state, crystallographic orientation and temperature. The cell state can be solid, liquid, interface, and re-solid. Solid state is assigned to all the cells at the beginning of the simulation, and may change to liquid, interface and then re-solid according to specific rules, as will be introduced later.

The initial solid cells belong to different grains. Each grain has its unique crystallographic orientation, and solid cells with the same crystallographic orientation combine to reconstruct the shape of this grain. For the AISI304 stainless steel to be investigated in this work, the dominating phase in the substrate as well as in the weld zone is Austenite with the crystal lattice structure of face-centered-cubic (FCC). The FCC grains, if growing in a uniform temperature field with no geometric constraint, should maintain an envelope of regular octahedron with equal growth preference along all the diagonal directions, as shown in Fig. 1a. The three diagonals combine to build a local 3D Cartesian coordinate for the grain (X' – Y' – Z'), and the angular relationship between this grain coordinate and the global coordinate of the entire computational domain (X – Y – Z) can be quantified by 3 Euler angles, Θ , Φ and Ψ , as graphically explained in Fig. 1b.

The temperature of each cell is calculated from the macro-scale model. Since the mesh size in the macro-scale model is much larger (30 μm) than the one in the current CA model (2 μm), a linear interpolation scheme is needed to map the temperature information onto the CA domain. As the molten pool passes by the CA domain, all the cells in the domain go through a heating–cooling cycle. If the local temperature goes above the liquidus temperature, the solid cells are changed to liquid, indicating the occurrence of melting, and the crystallographic orientations in these cells are erased. As the local temperature drops below the liquidus temperature, the liquid cells on the fusion line will be changed to interface cells, and the crystallographic orientations in these interface cells are inherited from the closest solid neighbors in the un-melted substrate. The re-solidification of the molten material will start from these seed interface cells on the fusion line.

The 3D growth of each interface cell is numerically tracked based on the de-centered square method [40]. In this method, each interface cell is assumed to expand with an envelope of regular octahedron, and the envelope is rotated with respect to the global coordinate according to the three Euler angles so that the envelope diagonal is well aligned with its crystallographic orientation. An

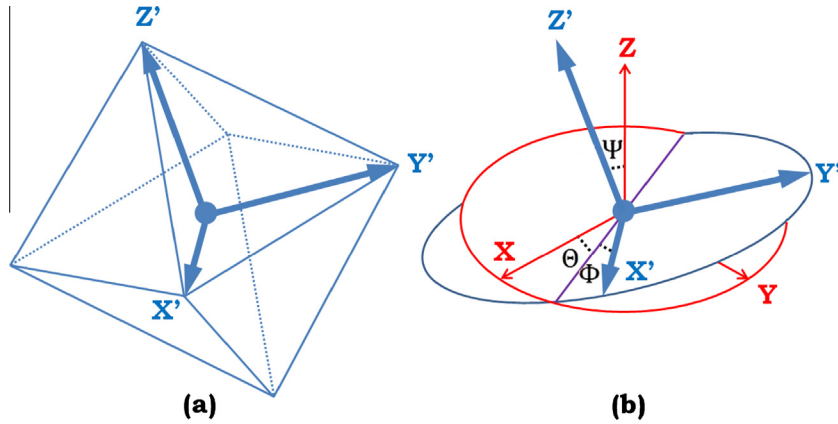


Fig. 1. Grain shape and crystallographic orientation in 3D Cellular Automata model: (a) grain grows with octahedron shape with equal preference along all diagonal directions; (b) the crystallographic orientation of a grain with respect to the global coordinate is defined with three Euler angles.

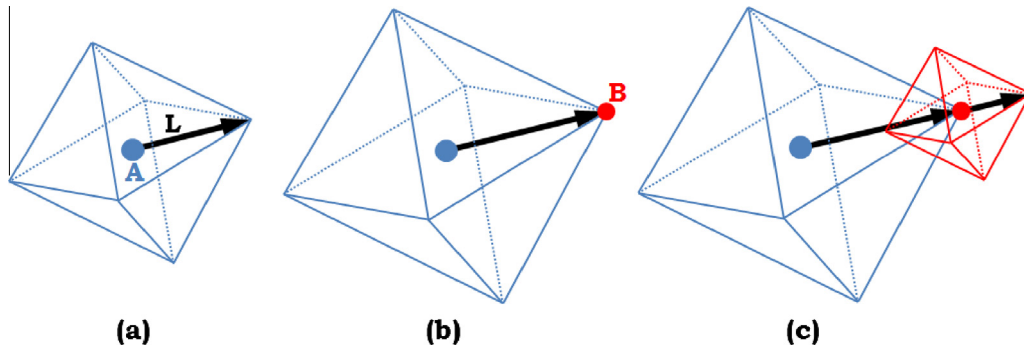


Fig. 2. Schematic illustration of 3D De-center square algorithm: (a) parent cell expand its octahedron envelope; (b) parent envelope capture new interface cell; (c) new interface cell expands its own octahedron envelope.

example interface cell (cell A) with its growth envelope is shown in Fig. 2a, and the half length of its envelope diagonal is defined as L . This term can be calculated by integrating the grain growth velocity over time, as shown in Eq. (14).

$$L = \sum_t V_l \Delta t \quad (14)$$

Here V_l is the grain growth velocity, which is assumed to be a pure function of temperature [28] and is determined as given in Appendix A; and Δt is the time step of the model. Given sufficient time, the envelope will grow out of the cell boundary to touch the neighboring cells. The cell boundaries are not shown in Fig. 2 so that the boundary of the growth envelope can be better visualized. The touched cells, if being liquid, will be captured as a new interface cell. One example of the new interface cells is denoted by the red¹ dot (cell B) in Fig. 2b. This new interface cell will inherit the crystallographic orientation from the “parent” cell and will have its envelope centered at the corner of the “parent” envelope. Note that the envelope center does not necessarily overlap with the cell center. The new interface cell will expand its envelope (denoted by the red octahedron in Fig. 2c) with its own growth velocity, and capture other liquid cells to be interface cells as the envelope expands large enough. The “parent” cell, if totally surrounded by

other interface cells, will change its cell state to be re-solid, indicating the complete of solidification of the cell.

According to the above algorithms, some seed interface cells with inherited crystallographic orientations are generated at the fusion line of the molten pool, and they will start a chain process to get more and more liquid cells captured to become interface and then re-solid cells. Each liquid cell, at a certain time, will be captured by an interface cell and inherit a specific crystallographic orientation, with which it will grow until becoming re-solid. Finally, all the melted cells will become re-solid cells with specific crystallographic orientations. The re-solid cells with the same orientation reconstruct the shape of a grain, thus the shapes of all the grains and their distribution in the re-solidified weld zone can be predicted.

2.3. Two-dimensional micro-scale dendrite growth model based on Cellular Automata and Phase Field methods

While the 3D distribution of the grains can be predicted by the meso-scale CA model, the microstructure development on the sub-grain level is still unknown. The microstructure within the grains is mostly in the form of dendrite, and the dynamic growth of the dendrite can be simulated by a Cellular Automata–Phase Field (CA–PF) model. In this integrated model, a CA model is built to track the dendritic morphology evolution and the associated solute redistribution, while the PF model is built to determine the growth kinetics for the CA interface cells, including their interface velocities and

¹ For interpretation of color in Figs. 2, 4 and 5, the reader is referred to the web version of this article.

solal concentrations on the interface. The major information of the model will be given below, while an exhaustive description can be found in [29–31]. Since the molten pool is huge compared with the dendrites therein, 3D simulations of the dendrite growth in the entire molten pool can cause unacceptably high computational expense. Therefore, the CA–PF model is now developed in a 2D format to predict the dendrite morphology on certain planes in the molten pool. It has been demonstrated in [20,22,24,26,29–31] that 2D models can produce quantitative predictions as long as the dendrites growth is directional and the growth direction is contained in 2D planes.

2.3.1. Cellular Automata component

In the CA component, a 2D computational domain is built and discretized into a finite number of square cells. Each cell carries the attribute variables of cell state (solid, liquid, interface or re-solid), concentration C_i and temperature T . Extra variables, including the interface normal direction β , interface curvature κ and crystallographic orientation θ , are defined for the cells with the state being “interface”. Among all the above variables, the temperature is determined from the simulation result of the macro-scale model using a linear interpolation scheme, the interface normal and curvature are calculated by the auxiliary algorithms described in [29], and the crystallographic orientation θ of different dendrites is assigned randomly at the fusion line.

The evolution of the cell state is governed by a 2D version of the de-centered square method, which has been well described in [29–31]. The method is conceptually identical to the 3D version (in Section 2.2) with differences in several minor issues. Specifically, the growth envelope is assumed to be square in the 2D algorithm (instead of octahedron in the 3D version), and the crystallographic orientation can be quantified by only one angular variable, θ (instead of three variables in the 3D version). The solid fraction of an interface cell should be calculated according to Eq. (15),

$$f_s = \frac{L}{\Delta x(|\sin\theta| + \cos\theta)} \quad (15)$$

in which L is the half diagonal length of the 2D envelope with square shape square envelope, θ is the 2D crystallographic orientation, and Δx is the mesh size of the 2D-CA computation domain. As f_s of an interface cell increases to above 1, the interface cell is changed to be re-solid. It is noted that the L in Eq. (15) is calculated also with Eq. (14), but the interface velocity is a function of local temperature and solal concentration. The dependence is yet to be found by the PF component.

The solal redistribution is calculated in the micro-scale CA model. This is the key difference between the micro-scale CA model and its meso-scale counterpart. The solal concentration field in and around the dendrites can be modified by two mechanisms, i.e. the solute partition at the interface and the solal diffusion in both the solid and liquid phases. Both mechanisms have been incorporated into Eq. (16).

$$\frac{\partial C_{eq,i}}{\partial t} = \nabla \cdot (D_{eq,i} \nabla C_{eq,i}) + \frac{\partial f_s}{\partial t} (C_{liq,i}^l - C_{sol,i}^l) \quad (i = 1, 2) \quad (16)$$

Here $C_{eq,i}$ and $D_{eq,i}$ are the equivalent composition and diffusion coefficients, and $C_{liq,i}^l$ and $C_{sol,i}^l$ are the liquid and solid concentrations at the interface [29]. The current model is built for ternary alloys, and the subscript i indexes the two alloying elements, the concentrations of which are free variables. The second term on the RHS of the equation formulates the solute rejection from solid to liquid, and the interface concentrations terms of $C_{liq,i}^l$ and $C_{sol,i}^l$ are also expected to be found by the PF component.

2.3.2. Phase Field component

The dendrite interface, which is tracked by the micro-scale CA model, is considered as a series of circular segments with their distinctive interface curvatures. The growth kinetics of each of these circular segments can be simulated by a 1D PF model, which has been reformulated in the polar coordinate. The PF model calculates the temporal variation of the phase and concentration according to Eq. (17) [29].

$$\frac{1}{M_\phi} \frac{\partial \phi}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(\varepsilon^2 r \frac{\partial \phi}{\partial r} \right) - \frac{\partial f}{\partial \phi} \quad (17a)$$

$$\frac{\partial C_i}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \sum_j^n \left(D_{ij} \frac{\partial C_j}{\partial r} \right) \right) + \frac{1}{r} \frac{\partial}{\partial r} \left(r \sum_j^n \left(M_{ij} \frac{\partial^2 f}{\partial C_j \partial \phi} \right) \frac{\partial \phi}{\partial r} \right) \quad (17b)$$

Here ϕ is the order parameter of the Phase Field. While the pure solid and liquid regions have the order parameter to be 0 and 1, respectively, the interface is the region where $0 < \phi < 1$. M_ϕ is the phase mobility, r is the local radius, ε is the gradient energy coefficient, C_i is the concentration of the i -th component, and D_{ij} and M_{ij} are the (i, j) -th component of the material diffusivity and mobility matrices, respectively.

The calculation of the energy density function f is the basis of the governing equation. For pure liquid ($\phi = 0$) or solid ($\phi = 1$) phases, their energy density functions, f_l and f_s , can be calculated based on the sub-lattice model [41]. For the interface region ($0 < \phi < 1$), where the liquid and solid phases coexist, the energy density is calculated according to Eq. (18).

$$f(\phi, C_1, C_2, T) = g(\phi)W + f_s \left(C_{s1}^{equi}, C_{s2}^{equi}, T \right) + h(\phi) \left[f_l \left(C_{l1}^{equi}, C_{l2}^{equi}, T \right) - f_s \left(C_{s1}^{equi}, C_{s2}^{equi}, T \right) \right] \quad (18)$$

Here $g(\phi) = \phi^2(1 - \phi)^2$ is the double well function, W is the energy penalty for the interface, and $h(\phi) = \phi^3(6\phi^2 - 15\phi + 10)$ is the interpolating function. The equilibrium concentration of C_{s1}^{equi} , C_{s2}^{equi} , C_{l1}^{equi} and C_{l2}^{equi} are calculated based on the Kim–Kim–Suzuki model [42].

2.3.3. PF-based growth kinetics database for CA component

For ternary alloys, the local solidification condition can be quantified by 6 independent variables: temperature T , interface curvature κ , liquid concentration at interface C_{li}^l , and constant A_i ($i = 1, 2$). The two terms of C_{li}^l and A_i collectively describe the liquid concentration profile of the i -th component in front of the interface based on Eq. (19).

$$C_i(x) = C_{\infty,i} + \left(C_{li}^l - C_{\infty,i} \right) \exp(-A_i x) \quad (19)$$

These 6 variables should be given to the PF component for the setup of the initial condition, and the PF model can calculate the local interface growth, with the instantaneous growth kinetics extracted from the calculation result [29].

To save computational cost, the PF component is only applied to a number of sample problems with selected solidification conditions defined by the 6 variables, and the predicted growth kinetics for the sample problems are saved in corresponding 6D matrices. During a CA simulation, the local solidification conditions, i.e. the 6 variables, can be determined for each interface cell. The corresponding growth kinetics can then be found from the database with a linear interpolation scheme according to the solidification conditions. The details of the database development and interpolation are given in [30].

Table 1
Experiment parameters.

	Laser power (W)	Welding speed (m/min)	Aver. cooling rate (K/s)	Average dendrite arm spacing (μm)	
				Experiment	Simulation
1	1000	1	14,000	4.2	4.5
2	1000	2	22,000	2.7	3
3	1000	3	36,000	2.1	2.4
4	1000	4	54,000	1.6	1.8

3. Experimental procedure

A continuous wave fiber laser (IPG YLS-1000) is used for the experiments. The laser beam with the wavelength of $\lambda = 1070$ nm is focused on the top surface of the substrate in a spot of 0.24 mm diameter. While the laser power is maintained at 1000 W, the welding speed is increased from 1 m/min to 4 m/min, as shown in Table 1. When the welding process is complete, the weld beads are cut along different sections. The section samples, once prepared, are observed by an optical microscope for the weld geometry, an electron backscatter diffraction (EBSD) technique for the grain distribution and a Scanning Electron Microscope (SEM) for the dendrite morphology.

The 304 stainless steel plates are used as the substrate material, with its chemical composition given in Table 2. The physical properties of the materials, which will be used for the macro-scale keyhole model, are given in Table 3. The 304 stainless steel is simplified as a Fe–19Cr–9Ni (wt%) ternary system when calculating the dendrite growth. The Gibbs energy functions of different phases in this ternary system can be found in [41], and the diffusion coefficients of the alloying components are given in [43].

4. Result and discussion

4.1. Investigation of thermal history and solidification parameters

An example of the prediction results from the macro-scale keyhole and molten pool model is given in Fig. 3a. The keyhole wall is the iso-surface of Level-set value $\phi = 0$, which can be found from the solution of Level-set equation (Eq. (1)). To give better visualization of the keyhole shape, only a half of the condensed domain is shown and the non-condensed region is totally hidden. The boundary of the molten pool, which is approximated as the isothermal

surface of the solidus temperature (1697 K), is shown in Fig. 3b. A 3D Cartesian coordinate is defined in the figure, and the notation $[X, Y, Z]$ is used to indicate 3D vector directions within this coordinate. The laser travels along the +X direction, which can be written as $[X = 1, Y = 0, Z = 0]$ or $[1, 0, 0]$; and the laser is delivered along the –Y direction, i.e. $[X = 0, Y = -1, Z = 0]$ or $[0, -1, 0]$, to the substrate. The top surface of the substrate is positioned at $Y = 0$, and the center of the laser is always located at $Z = 0$. This coordinate system will be used throughout the discussion in this paper.

If the 3D surface of the pool boundary is projected onto a Y–Z plane, the cross-sectional shape of the weld can be found. The predicted results are shown in Fig. 4. Only the right halves of the predictions are shown in the figure and compared with the left halves of the experimental results. The predicted boundary of the weld zone, i.e. the fusion line, is marked out by the interface between the pure blue region (indicating unmelted substrate) and the region of other colors. The colors other than blue denote the magnitude of cooling rate during solidification, and its calculation will be introduced later in this section. Good agreements have been obtained between the predicted cross-sectional weld shapes and the experimental observation, suggesting that the macro-scale model is giving reasonable prediction of the dynamic molten pool.

The temperature in the molten pool, especially in the mushy region, has decisive effects on the solidification process. The local temporal and spatial variations of the temperature are usually quantified by 4 solidification parameters, namely the temperature gradient G , growth rate R , the cooling rate $CR = G \times R$, and the factor $F = G/R$. The growth rate R is the advancing velocity of the solidification front towards its normal direction. This term is highly related to the welding speed V , as shown in Eq. (20),

$$R = V \cos \beta \quad (20)$$

where V is the welding speed, and β is the angle between the laser scanning direction and the normal direction of the solidification front. The F factor determines the morphology of the microstructure. As the term decreases, the potential microstructure morphology tends to change from planar, cellular, columnar dendrite to equi-axed dendrite.

In the present study, the solidification parameters of the molten pool can be readily calculated from the macro-scale model prediction, and the calculation results for a weld-cross-section of case 2 is given in Fig. 5. Again, only the right halves of the prediction results are shown in the figure. The weld center line is located on the left edge of each sub-figure, and the fusion line is marked out by the interface between the pure blue region and the region of other color. For each point on the cross-section, the parameters are calculated at the moment when its local temperature goes from above to below the liquidus temperature, i.e. when solidification starts. It is noted that different points on the cross-section start solidification at different times: the region near the fusion line always solidifies before the region near the weld center line. There is no solidification and hence the solidification parameters are defined

Table 2
Chemical composition of 304 stainless steel (wt%).

	C	Mn	Si	Ni	Cr	Mo
AISI304	0.08	2.0	1.0	9.0	19.0	~0

Table 3
Physical properties of 304 stainless steel and Argon gas [8,13,14,44–47].

Property	304 stainless steel			Argon
	Solid	Liquid	Vapor	
Density (kg m^{-3})	7200	6900	Ideal gas	Ideal gas
Phase change temperature (K)	1697 (solidus)	1727 (liquidus)	3375 (boiling)	
Specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)	712	837	495	520
Phase change latent heat (J kg^{-1})		2.74E5 (melting)	6.36E6 (evaporation)	
Thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)	19.2	209.2	T dependent	T dependent
Viscosity ($\text{kg m}^{-1} \text{s}^{-1}$)		6E–2	T dependent	T dependent
Surface tension coefficient ($\text{N m}^{-1} \text{K}^{-1}$)		–0.43E–3		
Thermal expansion coefficient (K^{-1})		1.96E–5		

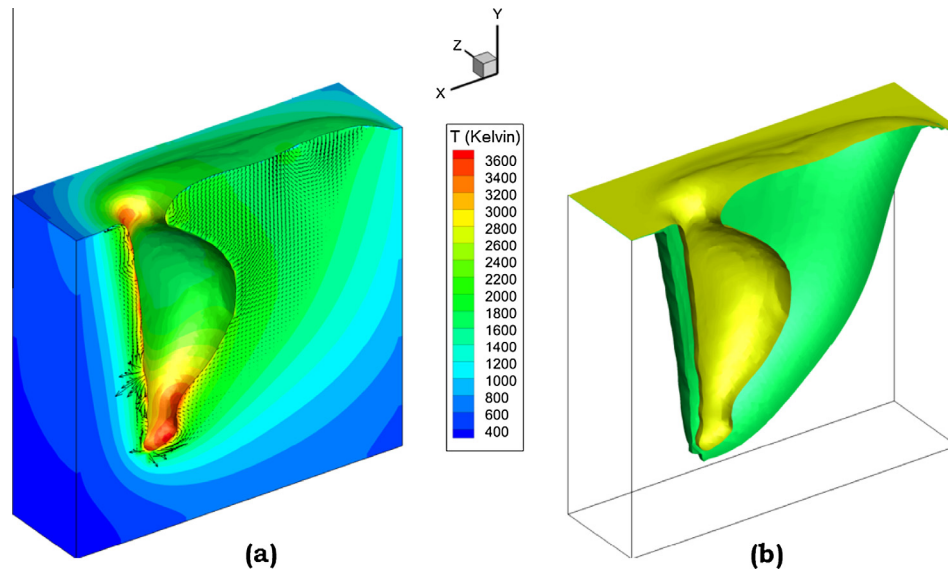


Fig. 3. Macro-scale modeling results in 3D Cartesian coordinate: (a) predicted temperature field in molten pool; (b) predicted shape of keyhole and molten pool.

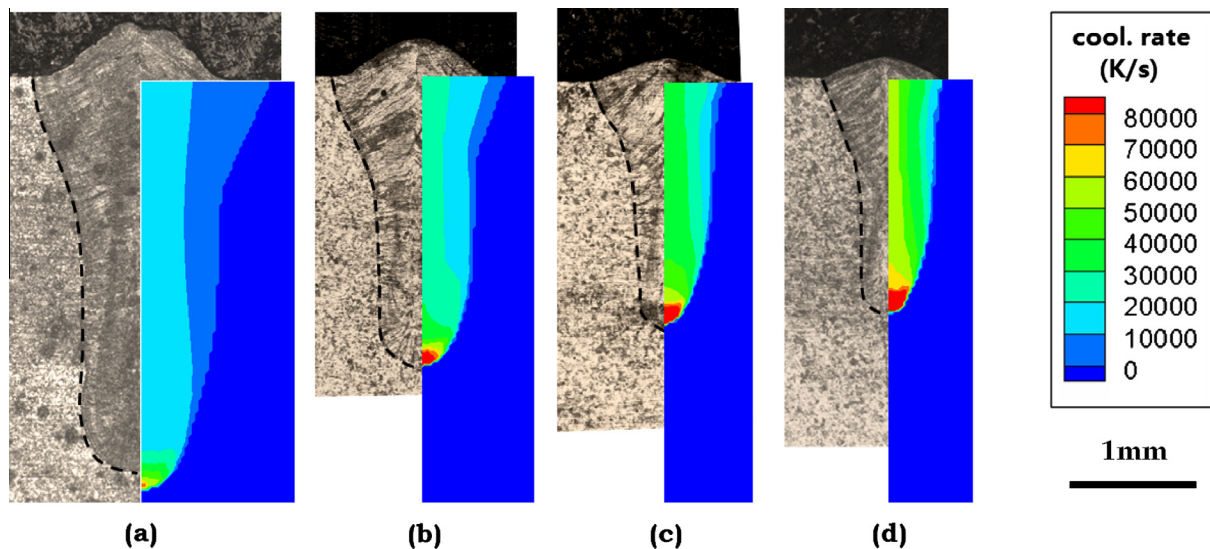


Fig. 4. Predicted weld cross-section and distribution of cooling rate for: (a) case 1, (b) case 2, (c) case 3 and (d) case 4.

to be zero in the un-melted substrate, which is colored by pure blue.

The temperature gradient G is found to be the highest at the bottom of the molten pool, where the heat sink effect of the substrate is most significant; and the gradient term is gradually decreased as the solidification advances towards the top of the weld zone near its center line. The growth rate R is found to be low near the fusion line, because when this region solidifies, the normal direction of the local solidification front is mostly aligned with $[0, 0, \pm 1]$ and hence perpendicular to the laser scanning direction of $[1, 0, 0]$. This makes $\beta \rightarrow \pi/2$ and $R \rightarrow 0$ in Eq. (20), leading to a very small value for R . On the other hand, the region near the center line solidifies last at the time when the pool tail passes this cross-section. The solidification front at the pool tail should be well aligned with the laser scanning direction, which gives $\beta \rightarrow 0$ and $R \rightarrow V$ according to Eq. (20). With the given value of G and R , the cooling rate CR and morphology factor F can be calculated. It is shown that the cooling rate is generally higher near the weld center line. This suggests the microstructure be finer near the weld

center line than in other regions in the weld zone. The morphology factor F maintains a high value near the fusion line, indicating the potential formation of cellular or even planar microstructure; the value gradually decreases towards the weld center line, and columnar or even equi-axed dendrites can be expected near the center line of the weld zone.

In the other 3 cases, the variation trends of the solidification parameters on the cross-section are qualitatively similar to those in case 2, but quantitatively the magnitude of each term is different from case to case. For example, the magnitude of cooling rate of all the cases can be compared in Fig. 4. As the welding speed is changed from 1 m/min to 4 m/min, the average cooling rate on the cross-section is increased from 14,000 K/s to 54,000 K/s.

4.2. Investigation of grain growth in laser keyhole welds

The meso-scale 3D CA model can be used to simulate the competitive grain growth during solidification and predict the final grain distributions, and an example of the 3D prediction results

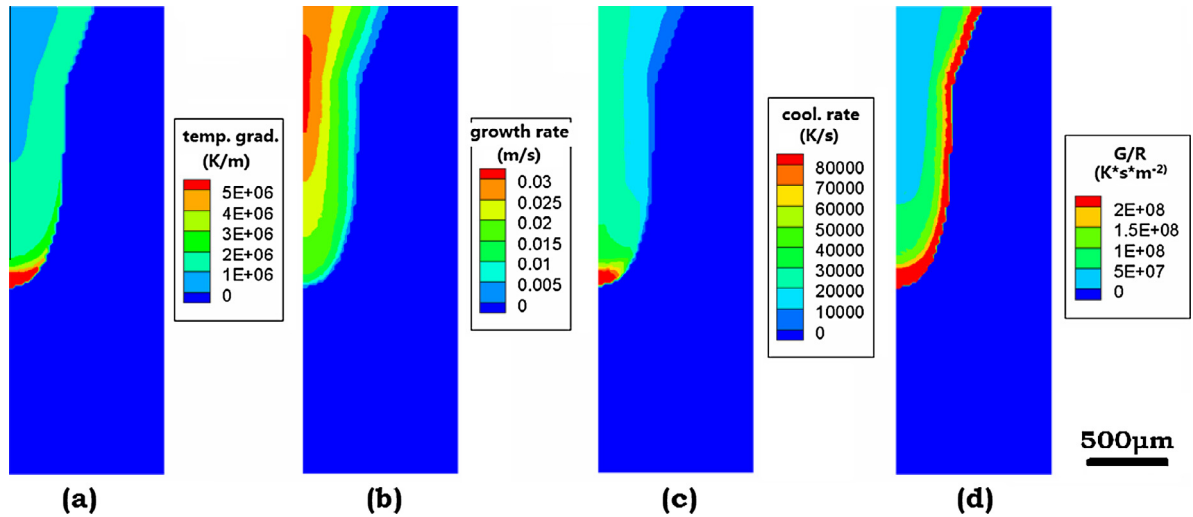


Fig. 5. Distribution of solidification parameters on cross-section of case 2: (a) temperature gradient, (b) growth rate, (c) cooling rate and (d) morphology factor.

is given in Fig. 6. In this example case, a keyhole-induced molten pool travels from the right to the left on a substrate, and a portion of the substrate is melted and re-solidified to produce a new grain structure in the weld zone. The profiles of the keyhole and molten pool are given on the left side of the figure, and the produced weld zone in the substrate is shown on the right side. The grains are colored differently in the figure according to the 3rd angular variable (Ψ), and the boundary of the weld zone is marked out by the dashed lines. While grains of equi-axed shape occupy the unmelted substrate, the weld zone is dominated by elongated grains that grow from the fusion line towards the weld center plane.

Since the 3D result is difficult to be visualized, several 2D sections along different directions are extracted from the 3D result and compared against the EBSD results.

One cut is made along the cross-section (parallel to the Y–Z plane) of the weld, on which the weld zone resembles a “wine-cup” shape, as shown in Fig. 7a. Two zoom-in images of the predicted result are also given, one for the “cup-cone” region (Fig. 7b) and another for the “cup-leg” region (Fig. 7d). Only the grain boundaries are given for the prediction results, and they show fairly similar grain distribution patterns to the EBSD results

(Fig. 7c and e). The results show the weld being dominated by grains of elongated shape, which are exclusively composed of columnar dendrites that spur from the fusion line and stretch towards the weld center line. Some grains of equi-axed shape can also be found near the weld center line, and there is a possibility for these equi-axed grains to be composed of equi-axed dendrites, as suggested by the low value of morphology factor F in Fig. 5. However, an SEM image given in Fig. 7f for the region in the “cup-cone” near its center line, where the equi-axed grains have been found, shows few equi-axed dendrites but prevailing columnar dendrites. Moreover, while all the columnar dendrite arms grow towards the up-right corner in the images, few of the arms can stretch across the entire image; instead, most of the arms only cover a certain distance before they are blocked by some other arms. This is because the growth direction of dendrite arms is not perfectly parallel to the SEM image plane, i.e. the cross-section plane, and most columnar dendrite arms will be intercepted by the cross-sectional plane at some point. Similarly, the elongated grains do not grow parallel to the cross-section. Some of these grains can be intercepted by the cross-sectional plane near the weld center line and appear to have equi-axed shape.

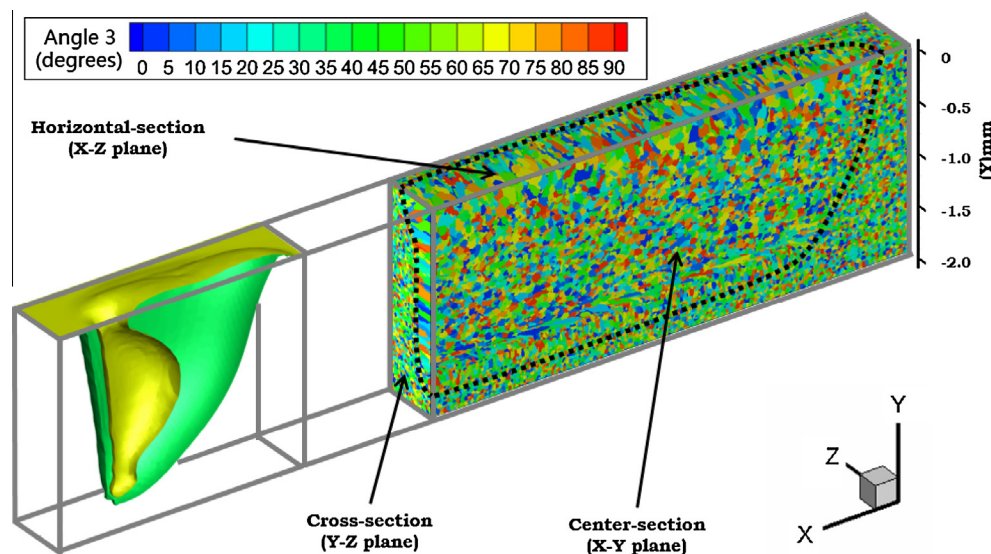


Fig. 6. Example result of 3D-CA model.

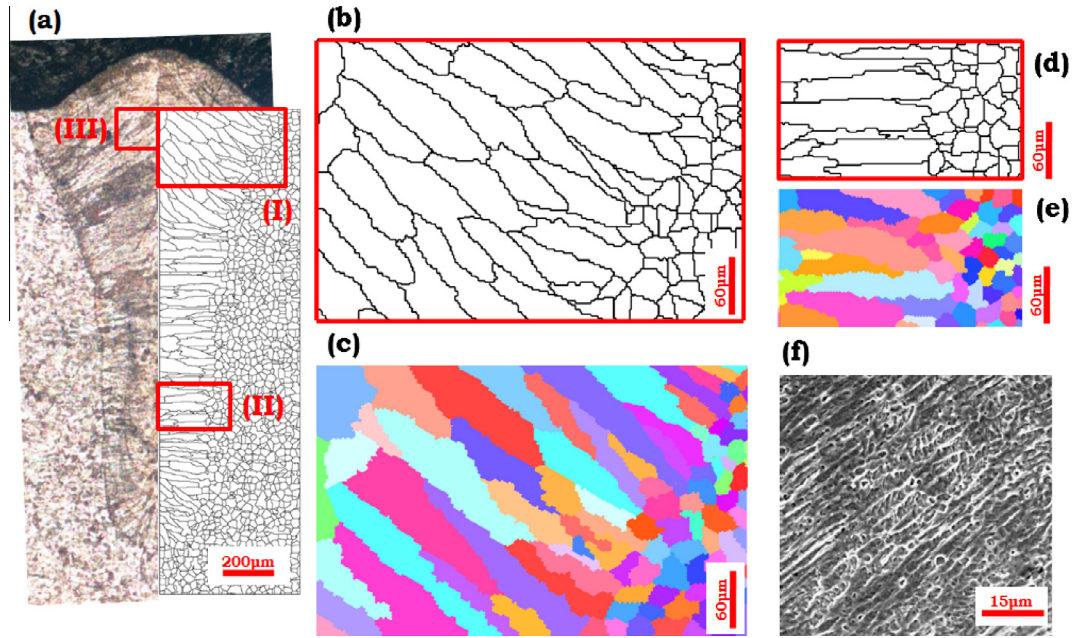


Fig. 7. Grain distribution on weld cross-section. (a) Entire view of cross-section; (b) 3D-CA prediction for region I; (c) experiment (EBSD) result for region I; (d) 3D-CA prediction for region II; (e) experiment (EBSD) result for region II; (f) SEM image for region III.

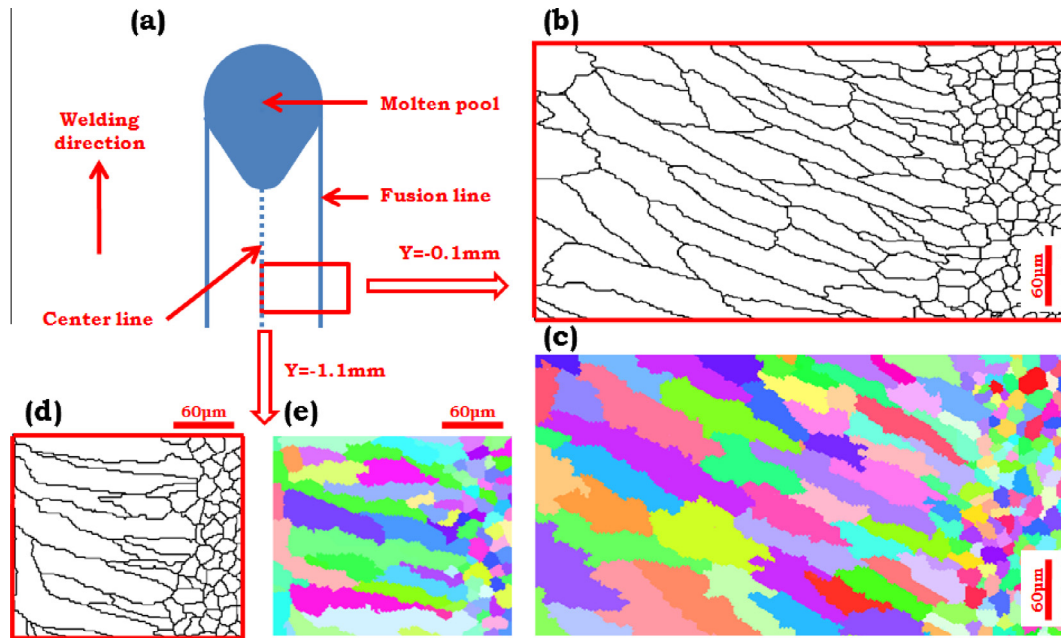


Fig. 8. Grain distribution on weld horizontal sections: (a) schematic view of horizontal section; (b) 3D-CA prediction for section of $Y = -0.1$ mm; (c) experiment (EBSD) result for section of $Y = -0.1$ mm; (d) 3D-CA prediction for section of $Y = -1.1$ mm; (e) experiment (EBSD) result for section of $Y = -1.1$ mm.

Two other cuts are made parallel to the X–Z plane, one at the depth of $Y = -0.1$ mm (right below the top surface of the substrate, horizontally cutting the “cup-cone”) and another one at the depth of $Y = -1.1$ mm (horizontally cutting the “cup-leg”). The predicted grain distributions on these sectional planes are compared with the EBSD scans in Fig. 8, which shows reasonable agreement. On the section of $Y = -0.1$ mm (Fig. 8b and c), elongated grains are prevailing near the fusion line, while equi-axed grains occupy the region near the weld center line. Again, these equi-axed grains

are the intercepts of elongated grains by the horizontal sections. On the section of $Y = -1.1$ mm (Fig. 8d and e), elongated grains dominate the entire section and few equi-axed grains can be found. This suggests that the growth direction of most elongated grains (and the columnar dendrites therein) are parallel to the horizontal plane. The grains start their growth almost perpendicularly with respect to the fusion line, with a little tilting towards the welding direction. However, as the molten pool almost passes by and the solidification gets close to the weld center line, the temperature

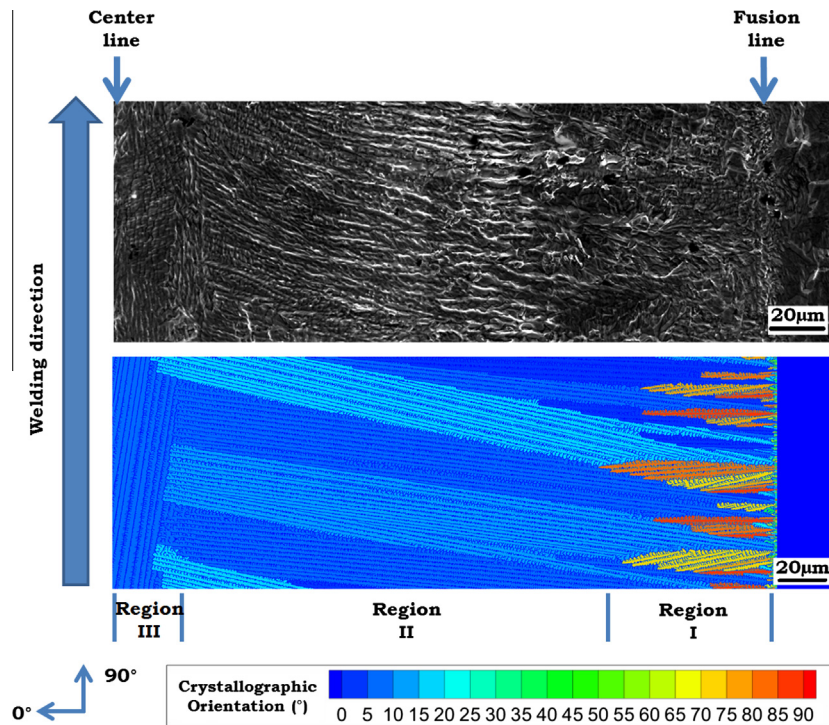


Fig. 9. Dendrite distribution in horizontal section of $Y = -1.1$ mm.

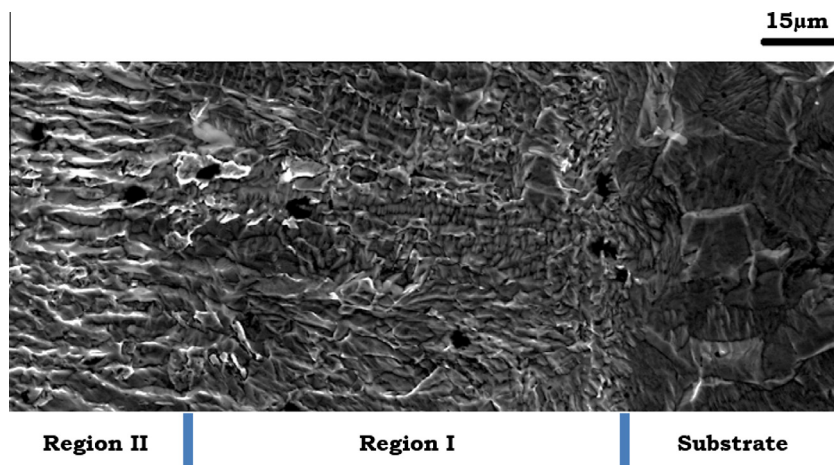


Fig. 10. Microstructure in region I is dominated by columnar dendrites with various crystallographic orientations.

gradient at the solidification front turns rapidly to be almost parallel to the welding direction, and the grains have to adapt themselves to this change by turning their growth direction towards the welding direction. This direction turning of grains can be observed in the EBSD result (Fig. 8e) and has been well replicated by the meso-scale model (Fig. 8d).

4.3. Investigation of sub-grain microstructure in laser keyhole welds

While the grain distribution can be predicted by the 3D meso-CA model, the dendrite morphology within the grains is still not known. CA–PF simulations are implemented to simulate the dynamic dendrite growth on the horizontal section at $Y = -1.1$ mm for all the 4 cases. As has been shown in Fig. 8d and e, the grains on this plane tend to grow horizontally, and

hence the dendrites therein are expected to grow also within the same plane. Given this scenario, the 2D CA–PF model should be able to provide quantitative predictions of the dendrite morphology.

The predicted microstructure in the right half of the horizontal section at $Y = -1.1$ mm in case 2 are shown in Fig. 9, in which the colors represent the crystallographic orientations of different dendrites. Since the dendrites are 4-folded symmetric, the crystallographic orientation value is selected between 0° and 90° . 0° is assumed to be perpendicular to the welding direction and 90° is assumed to be parallel with the welding direction. Note that the dendrites with 0° and 90° are actually identical. The predicted dendrite morphology is shown to be quite consistent with the SEM image, and three regions with different dendrite morphologies can be found from the fusion line to the center line.

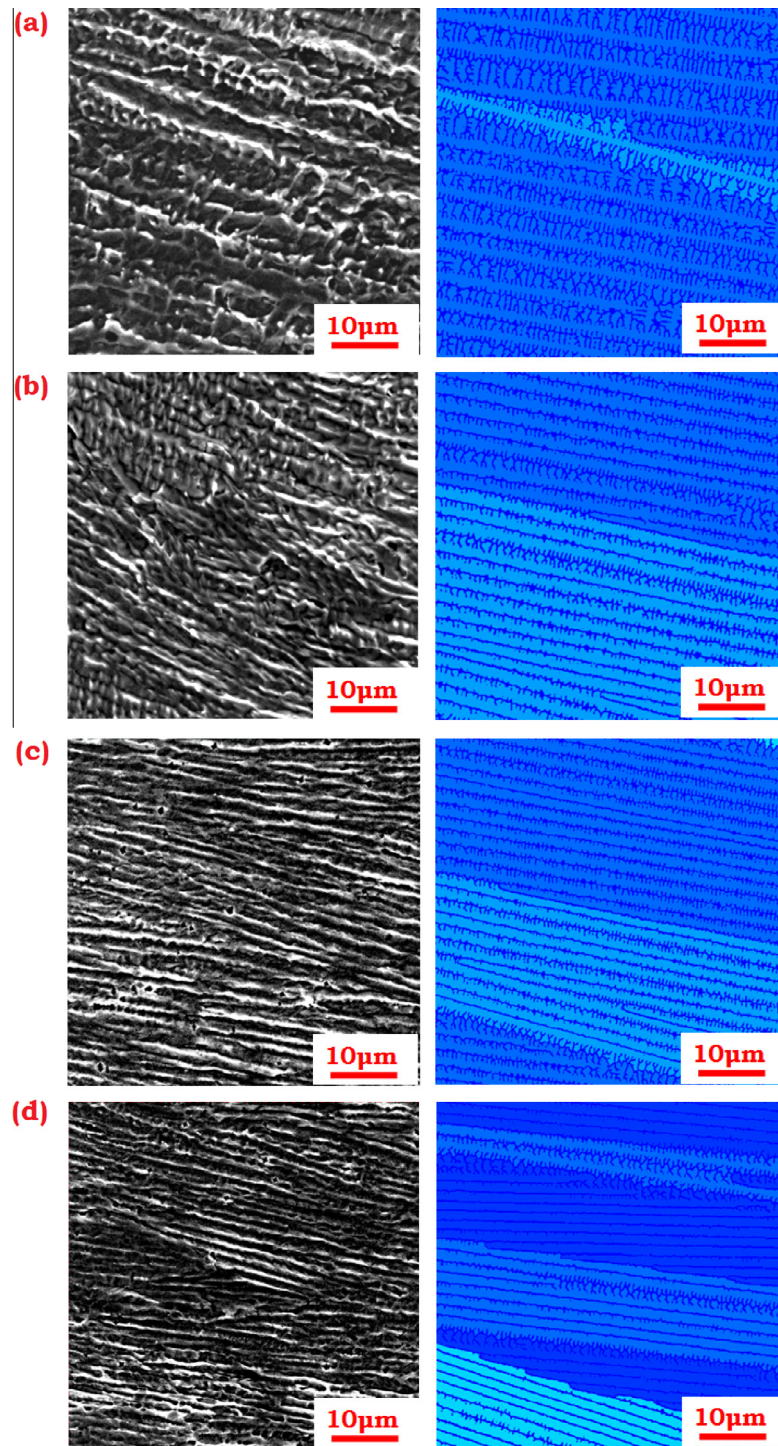


Fig. 11. Dendrite morphology in region II for (a) case 1, (b) case 2, (c) case 3 and (d) case 4.

4.3.1. Region I

In this region the dendrites inherit crystallographic orientations from the partially melted grains on the fusion line and start epitaxial growth. The distribution of the crystallographic orientation in the substrate can be approximated as random in the experiments, and hence random crystallographic orientations between 0° and 90° are assigned to the dendrites seeds on the fusion line in the simulation. The dendrites with various orientations all grow towards the weld center line, and hence the dendrites in region I are found to be quite disorderly. A high magnification SEM image of the microstructure in region I is given in Fig. 10, which clearly

shows columnar dendrites with diverse orientations coexisting in the region.

Fierce competitive growth takes place between different dendrites in region I, and the dendrites with their crystallographic orientations similar to the maximum temperature gradient direction are generally more favored to grow. The maximum temperature gradient is usually aligned with the normal direction of the molten pool, and in region I it is along the 10° direction. Therefore, the dendrites of about 10° orientations are favored most and can survive from region I and grow into region II.

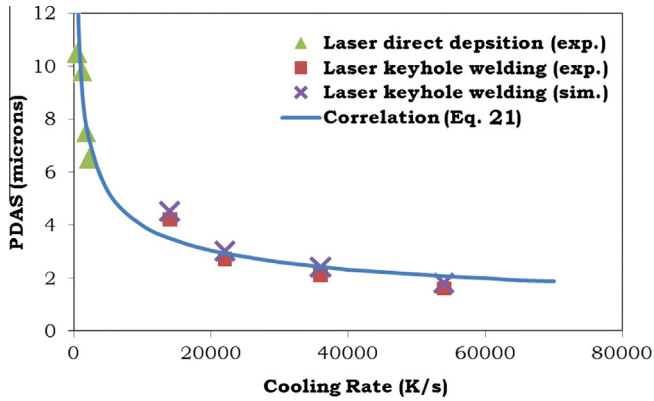


Fig. 12. Primary Dendrite Arm Spacing (PDAS) as a function of cooling rate.

4.3.2. Region II

Region II is totally occupied by the dendrites that survive from the competitive growth in region I. All the dendrites have very similar crystallographic orientations (around 10°), and hence the overall pattern of the dendrites is very uniform.

4.3.3. Region III

As the solidification front arrives at the center line, the normal direction of the molten pool is well aligned with the welding direction, i.e. $[1, 0, 0]$ direction. While this direction is perpendicular to the primary arms of those surviving dendrites, it matches the direction of their secondary arms. This greatly encourages the growth of the secondary arms along the welding direction, which is shown to dominate region III. It is noted that the turning of dendrite growth direction is consistent with the turning of grain growth presented in Fig. 8d and e.

While the dendrite growth direction has been well predicted by the model, the dendrite size is another huge concern for the welds properties and performance. The predicted dendrite morphology in region II for all the cases are graphically compared with the SEM images in Fig. 11. In case 1 with the low welding speed of 1 m/min, the primary dendrite arm spacing is rather coarse, and the secondary arms are also well developed. As the welding speed is increased, the primary dendrite arms become less sparse, and the secondary arm development is greatly suppressed. The variation trends of primary and secondary dendrites are well captured by the CA–PF modeling results. The Primary Dendrite Arm Spacing (PDAS) can easily be measured from the images, as given in Table 1, where quantitative agreement is found between the simulation and the experiment results. The PDAS is found to be decreasing from case 1 to case 4, which is mostly due to the higher cooling rate associated with the high welding speed (shown in Fig. 4). The measurement results of PDAS are drawn as a function of cooling rate in Fig. 12, together with the results from the laser direct deposition process in [30]. Combining the two sets of experiment data, a correlation between cooling rate and PDAS can be obtained by a curve-fitting technique, and the resultant correlation is given in Eq. (21).

$$\text{PDAS} = 144R^{-0.39} \quad (21)$$

5. Conclusion

In this paper, a multi-scale model has been presented to investigate the solidification of molten pools in laser keyhole welding processes. On the macro-scale, a multi-phase and multi-physics model is utilized to predict the thermal history of molten pool during solidification; on the meso-scale, a 3D Cellular Automata model

has been developed to predict the competitive grain growth; on the micro-scale, a numerical model integrating Cellular Automata and Phase Field methods has been utilized to simulate the dendrite growth.

The macro-scale model provides reasonable predictions for the solidification conditions, which is found to be profoundly affected by the welding speed. The cooling rate associated with laser keyhole welding can be above 10,000 K/s, and the solidification parameters vary significantly across the weld cross section.

The meso-scale model is used to reveal the 3D grain distribution in the keyhole welds, and the predictions are compared with EBSD 2D scanning results. The results show the dominance of elongated grains throughout the weld bead. The grains growth direction varies according to the complex shape of the keyhole-driven molten pool, and obvious competitions exist among different grains.

On the sub-grain level, the columnar dendrites prevail. The selection of the dendrite orientation is primarily controlled by the direction of maximum temperature gradient, while the dendrite morphology is significantly affected by the cooling rate. As the molten pool moves forward, the direction of the maximum temperature gradient, which is usually normal to the molten pool boundary, gradually changes from transverse to parallel with respect to the welding direction, and hence three different regions of different dendrite distribution can be identified for a horizontal section of the keyhole weld bead at its “wine-cup-leg” region. As the cooling rate is increased, the primary dendrite arms become less spaced, while the growth of secondary dendrite arms can be suppressed.

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Appendix A. Determination of temperature dependence for grain growth rate using CA–PF model

The CA–PF model has been used to simulate the growth of a single dendrite at different undercooling levels. The purpose of the simulations is twofold. The first is to validate the CA–PF prediction of stainless steel dendrite growth with analytical solutions, and the second is to find a correlation between the temperature and grain growth rate.

The simulation is implemented in a computational domain with 2000 by 2000 square meshes. The mesh size of $0.1 \mu\text{m}$ and the time step of $1\text{e}^{-7}\text{ s}$ are used for the simulations. The entire domain is kept at a specified temperature in the simulation, and the initial concentration of the entire domain is set to be 19 wt% Cr and 9 wt% Ni. A nucleation seed with the crystallographic orientation $\theta = 0^\circ$ is planted manually at the center of the domain, and the dendrite will grow self-consistently until its tip velocity reaches a steady value. The predicted morphologies at the end of the simulation for selected undercooling levels are shown in Fig. A1, and the predicted steady-state dendrite tip velocity is given in Fig. A2. The prediction is compared with the analytical solutions for Fe–18.5Cr–11Ni and Fe–18.5Cr–12.5Ni alloys in [48] and Fe–18Cr–8Ni alloy in [49], and a reasonable agreement has been achieved. A correlation is obtained by curve fitting to quantitatively describe the temperature dependence of dendrite tip velocity, as given in Eq. (A1).

$$V(\Delta T) = 7.325 \times 10^{-6} \times \Delta T^{3.12} \quad (\text{A1})$$

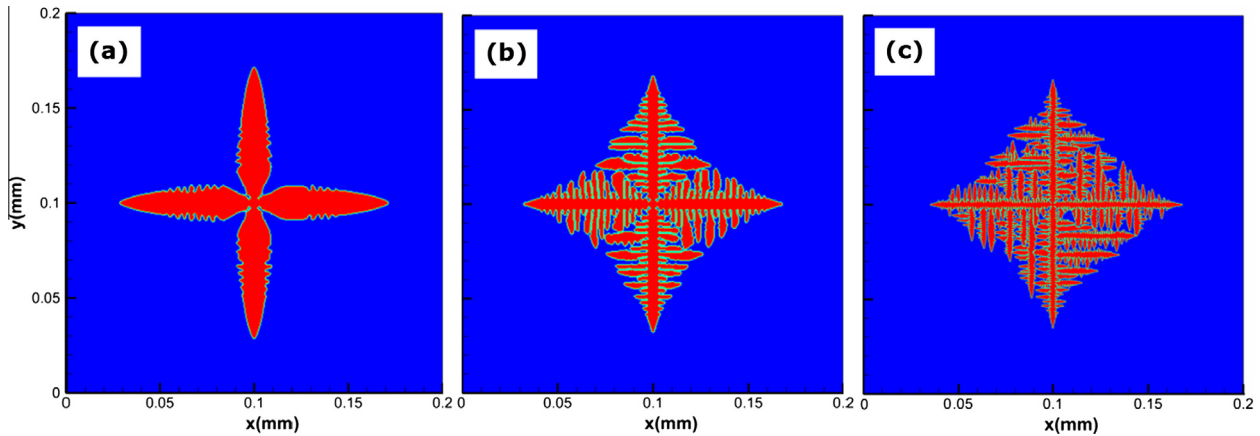


Fig. A1. Predicted dendrite morphology for (a) 5 K undercooling at 60 ms, (b) 15 K undercooling at 1.6 ms and (c) 25 K undercooling at 0.4 ms.

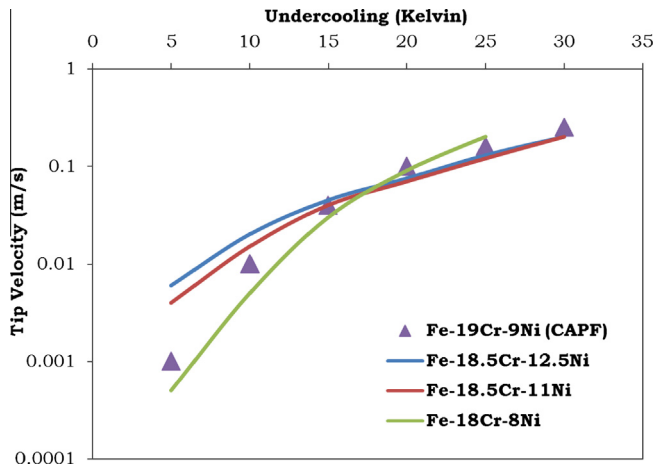


Fig. A2. Temperature dependence of dendrite tip velocity.

This quantity can be used as an approximation of the grain growth rate in the meso-scale model, and is used to calculate the V_I term in Eq. (14).

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